

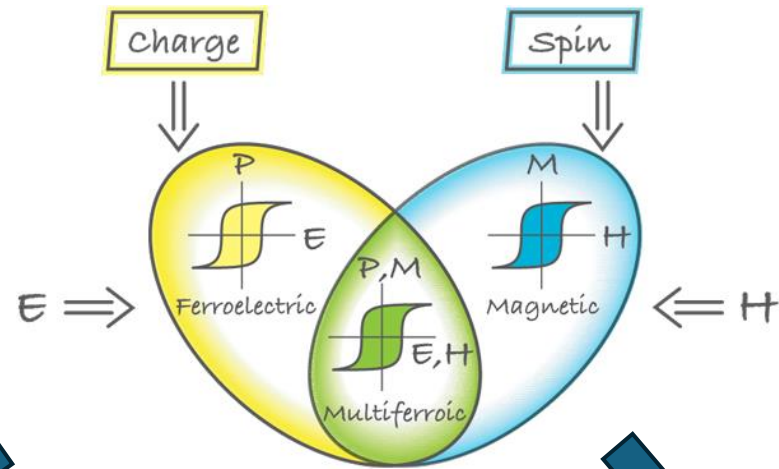
Magnetoelectric control in type-I multiferroics via domain walls

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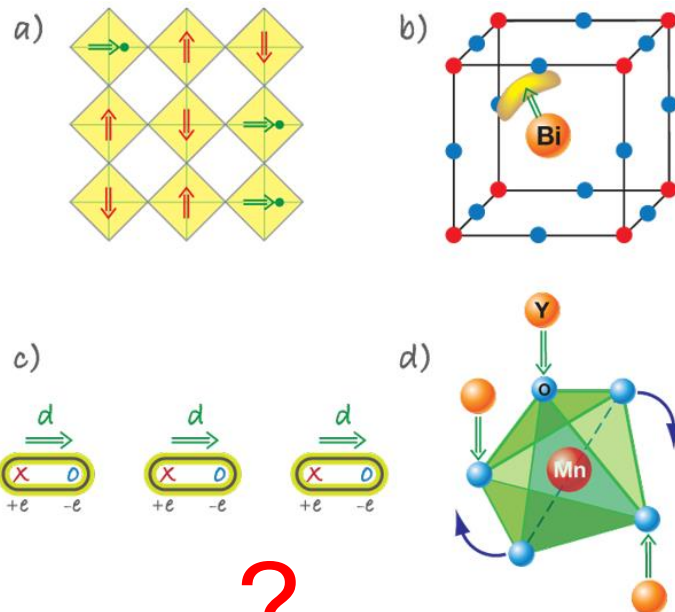
Multiferroics



Daniel Khomskii

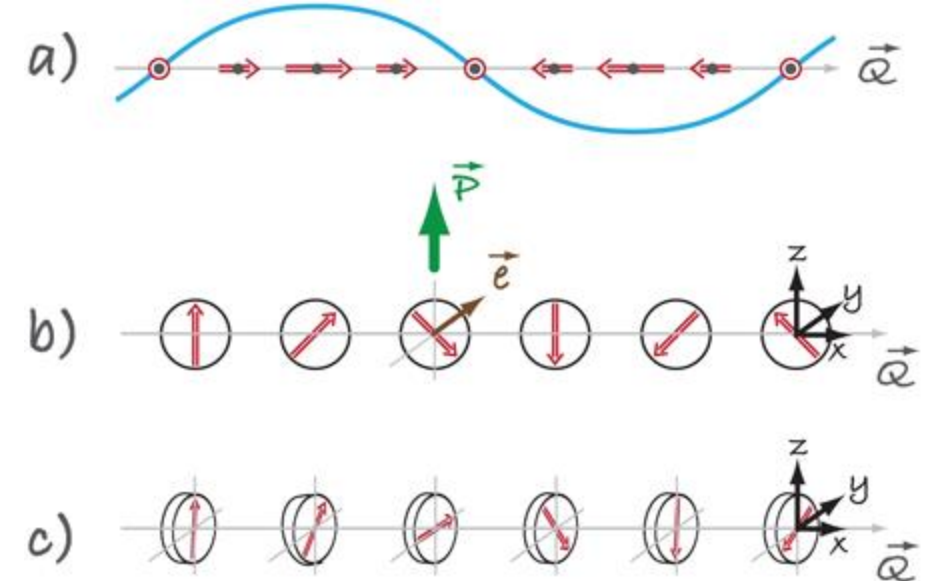
P on its own: Type I

P induced by spins: Type II



?

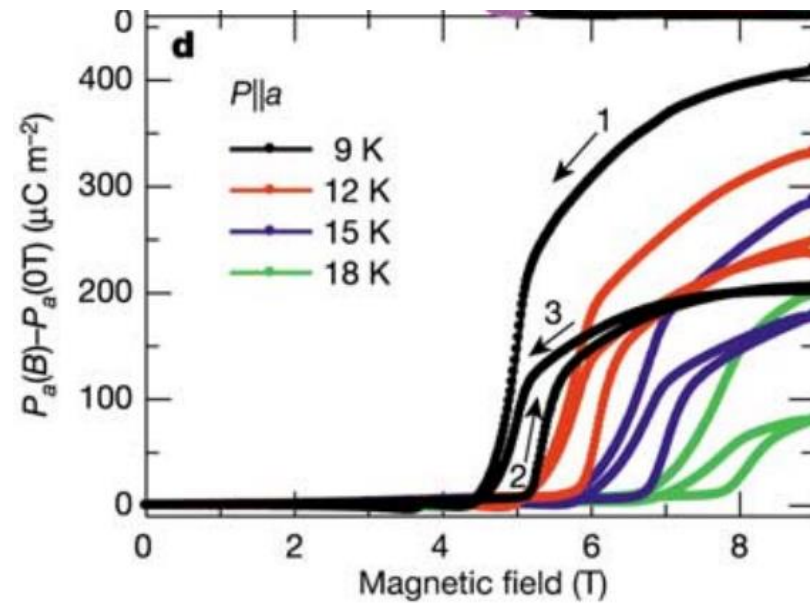
e.g: BiFeO_3



Strong ME control!

e.g: TbMnO_3

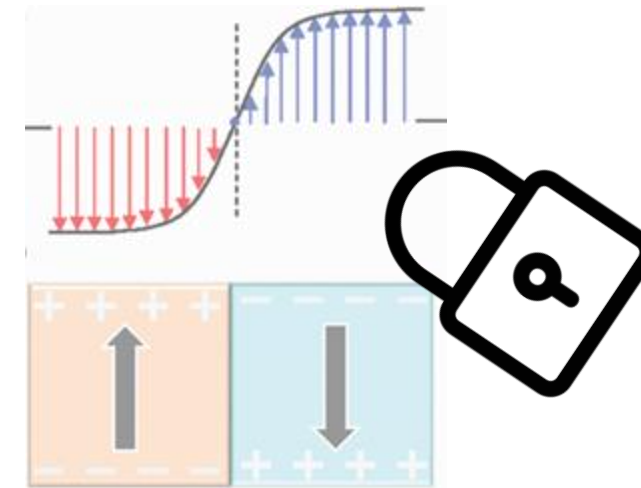
Strong ME control in type II



P flop triggered by B

T. Kimura et al., Nature 426, 55 (2003)

Strong ME control in type I?



Interlocking
& possible
Cross-
control?

D. M. Evans et al., Phys. Sci. Rev **5**, 67 (2020)

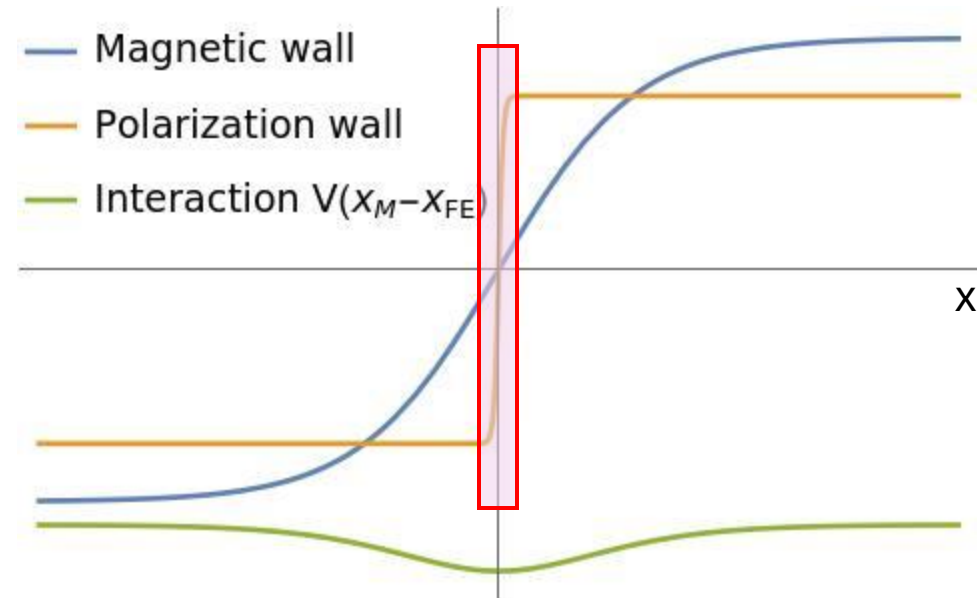
ME coupling via interlocked domain walls

- BiFeO₃, YMnO₃ do not display strong ME

- $F = \int d^3r [J(\nabla M)^2 - K_z M_z^2]$

- $E_{DW} \propto \sqrt{JK_z}$

- Nature of V

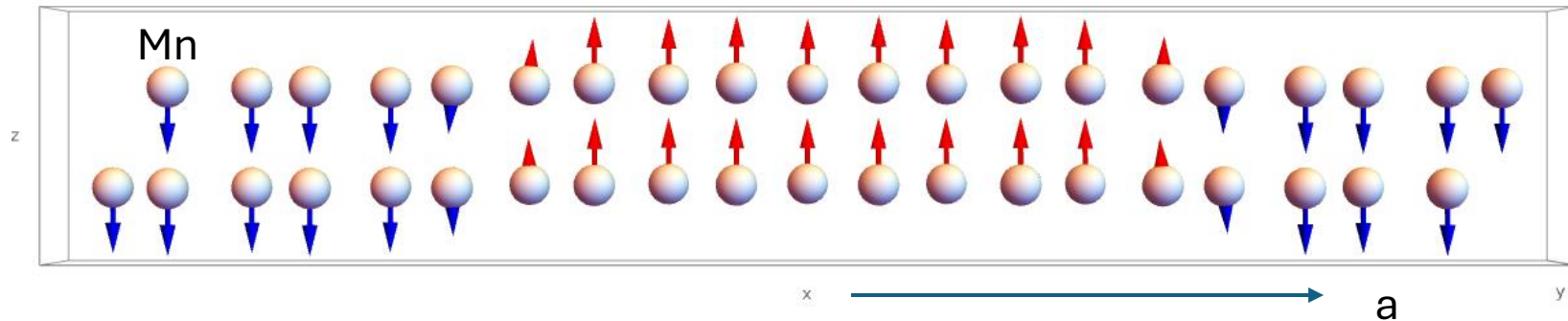
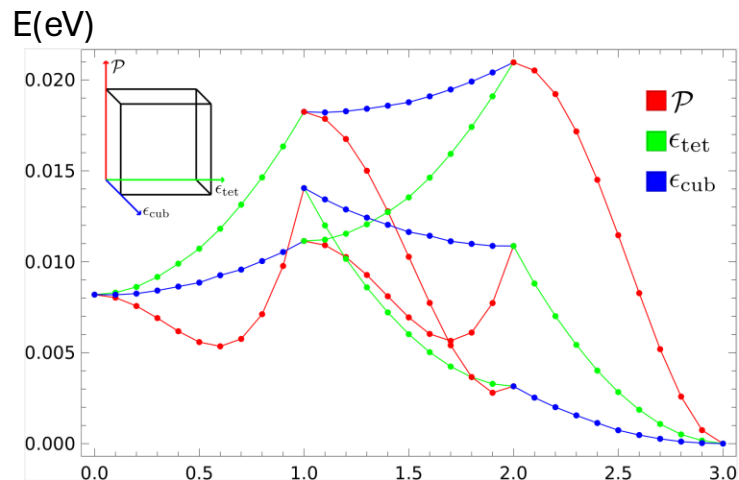
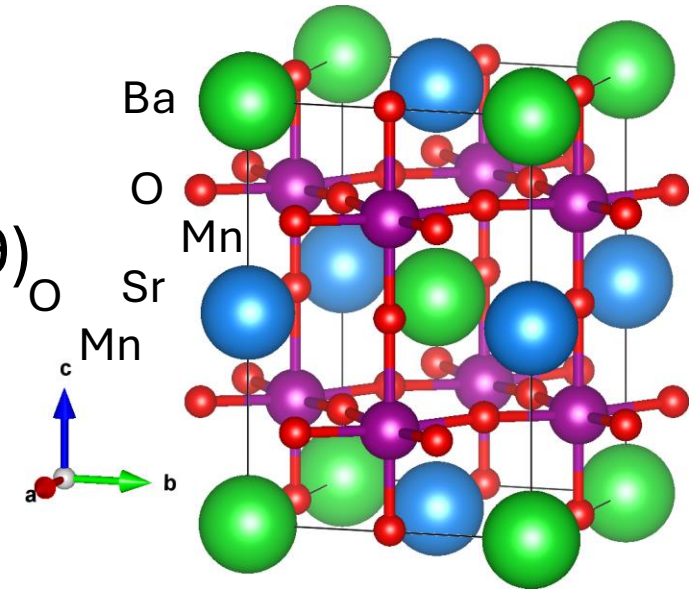


Methodology

- Freeze in the modes (ISODISTORT & LINVARIANT)
- Structural optimization with magnetism and Hubbard U
- Wannier tight binding Hamiltonian
- Heisenberg J from Katsnelson-Lichtenstein formula
 - Isodistort tool, Harold T. Stokes *et al.* stokes.byu.edu/iso/isodistort.php
 - Linvariant tool, Peng Chen *et al.* github.com/PaulChern/LINVARIANT

SrBaMn₂O₆

- Tetragonal structure, rocksalt order (Space gr: #99)
- G-type AFM order, $T_N \sim 185\text{K}$, $T_C = 400\text{K}$
- FE is suppressed by AFM order
- DFT: weak polar mode stabilization energy

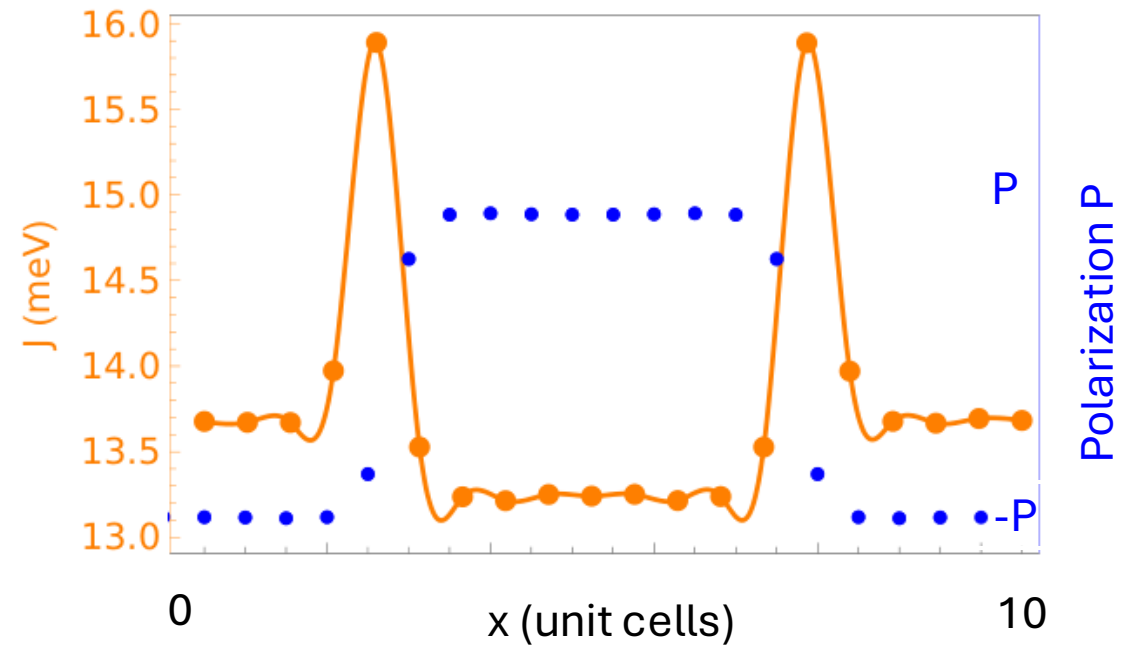


Exchange enhancement

- Magnetic Hamiltonian:

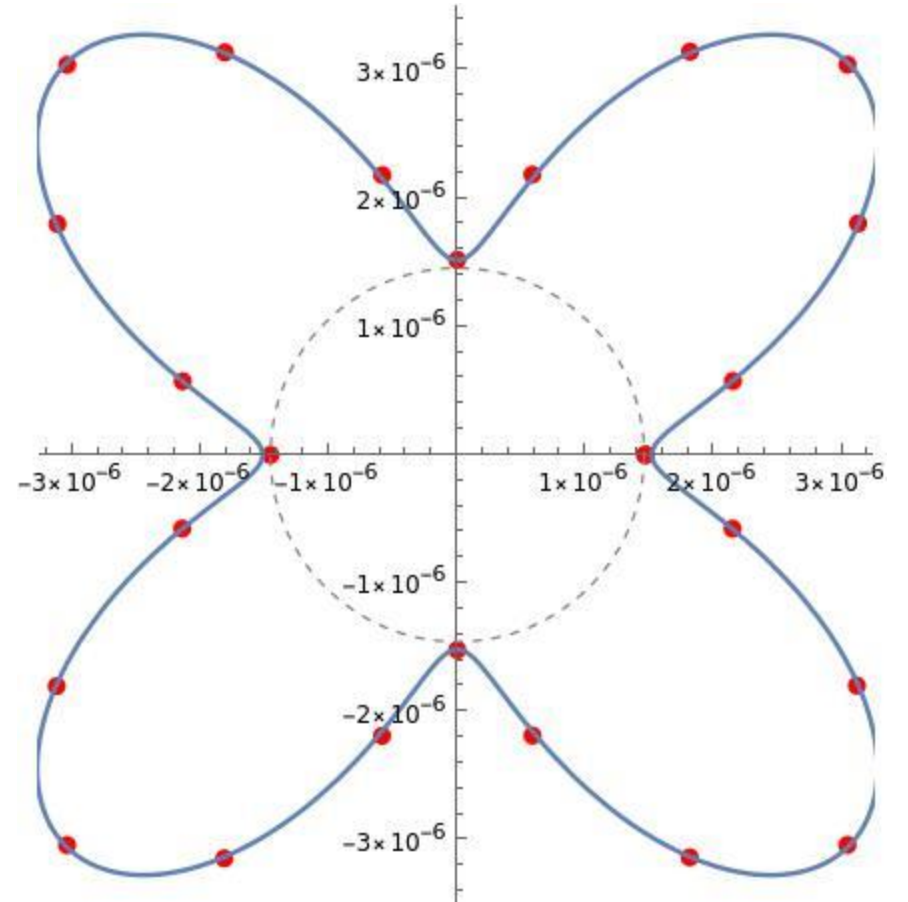
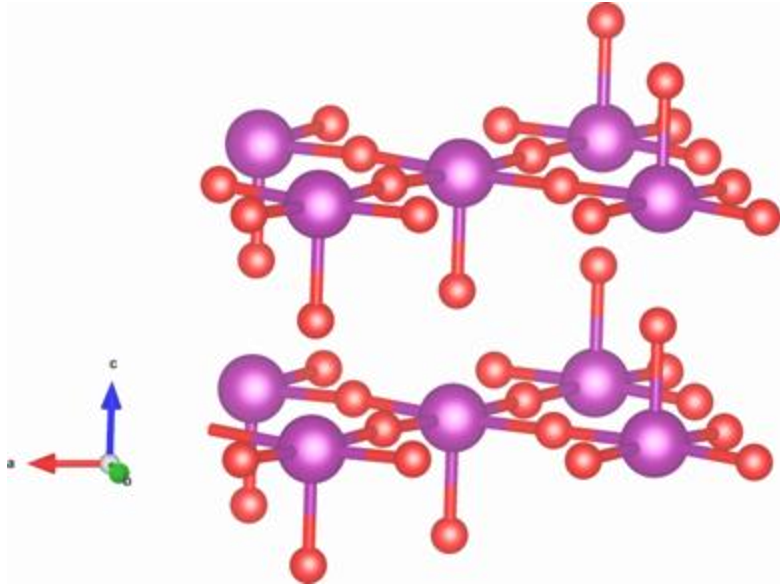
$$\mathcal{H} = -J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j + \sum_i (K_z S_{iz}^2 + K_{xy} (S_{ix}^4 + S_{iy}^4))$$

- Easy (XY) plane anisotropy
- Width of magnetic walls, $\lambda \propto \sqrt{\frac{J}{K}}$
- ~20% enhancement



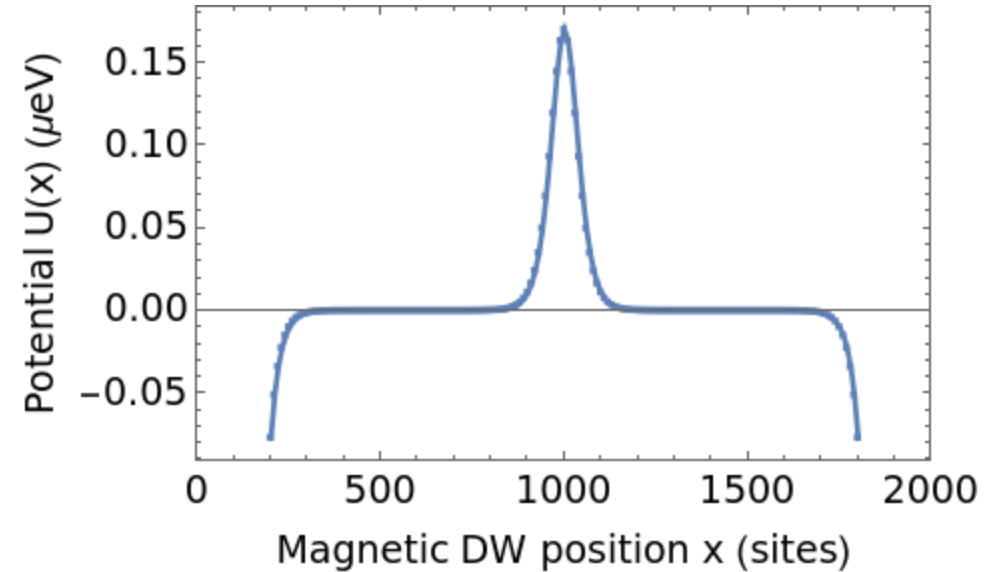
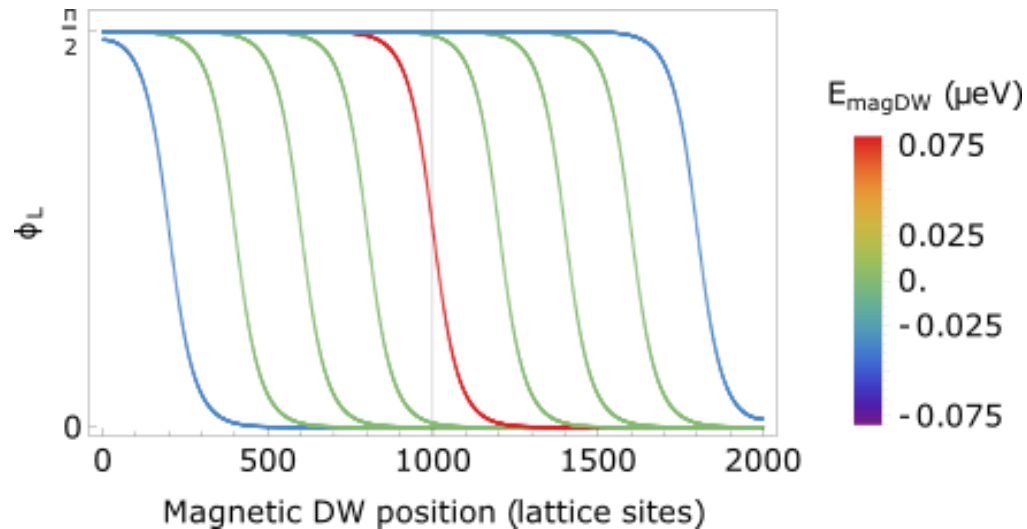
Magnetic Anisotropy & 90° magnetic DWs

- Hard z-axis $\rightarrow$ 4th order anisotropy
- $K_{xy}=1.4 \times 10^{-6}$ eV/Mn [DFT accuracy]
- Spins turn by 90° across DWs



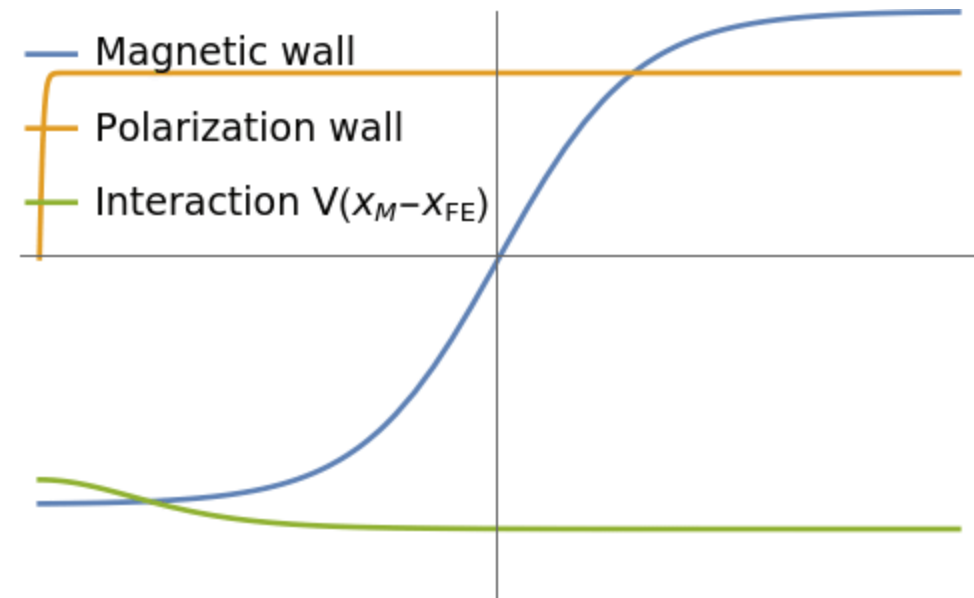
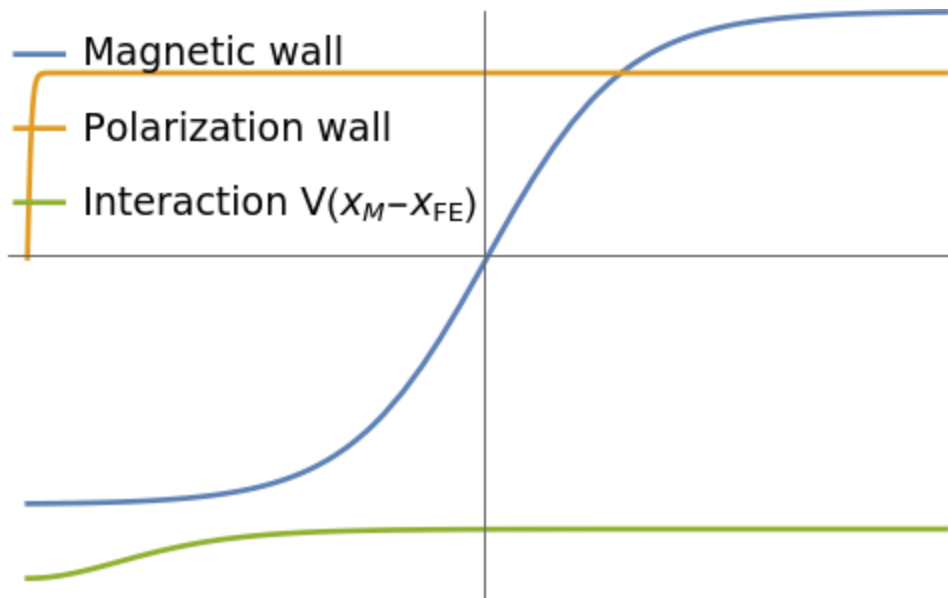
Computing DW interaction

- Position constrained
- Energy of DWs: $U(x)$



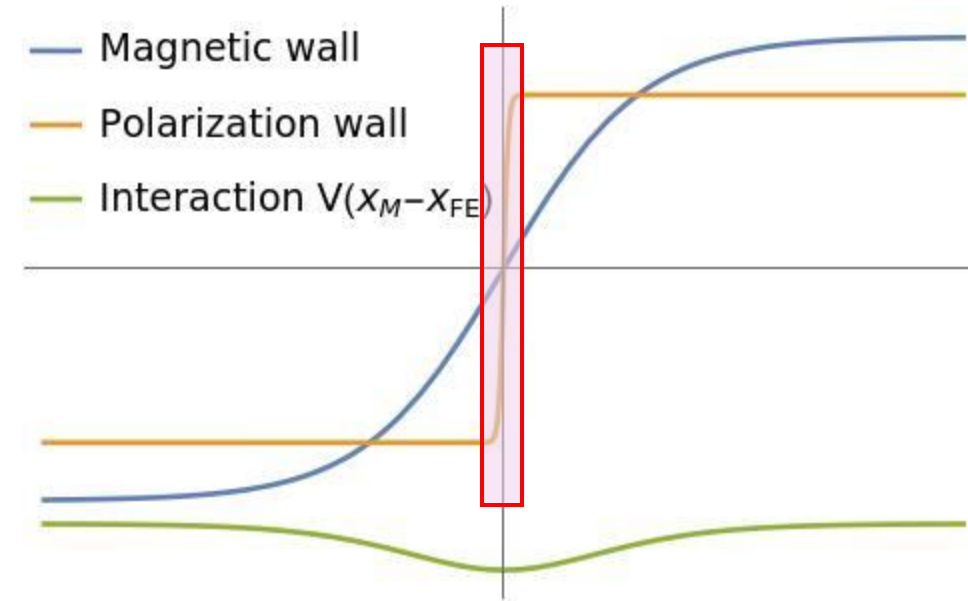
Interaction between magnetic and electric DWs

- At a FE DW:
 - J decreased: attractive interaction, clamping
 - J increased: repulsive interaction, FE DW can push but not pull



Conclusions

- Structural distortions at FE DW
 - Interactions between FE and magnetic DWs
 - Range: λ_M , K dependent
 - J increased $\rightarrow$ repulsive, can push
 - J decreased $\rightarrow$ attractive, clamping
- Clamped FE and magnetic DWs $\rightarrow$ ME control



Thank you!